Adobe Behaviour Simulation Challenge
Tweet Engagement Prediction and Content Generation
Team 40

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Abstract

In this report, we provide a thorough analysis of our methodology for two separate but related objectives in the framework of a tweet production and social media engagement prediction challenge in this study. Our first task, which involved simulating behaviour, required us to use OpenAI's CLIP model to extract textual descriptions from pictures and videos. But because of computational limitations, a different strategy was used, combining preprocessing methods including picture embedding and one-hot date-time attribute encoding, after which a unique Multilayer Perceptron (MLP) application was used to forecast tweet engagement. In order to generate meaningful tweet captions for content simulation, Task 2 required pretreatment techniques such as image extraction and thumbnail synthesis. The LLaVA model was then applied, along with few-shot prompting and prompt engineering. Our methods, which include preprocessing, strategic modifications, and machine learning models, are comprehensive and offer insights into the difficulties encountered and the processes involved in making decisions.

1 Background

Two main tasks in the behaviour simulation challenge are stated in the Adobe problem definition. Predicting user involvement (measured in likes) for tweets based on content, username, media URLs, and date is the first task. The second task is to create a tweet using the provided metadata, which includes the company, username, media URL, timestamp, and number of likes. There are 300K samples in the training dataset, which runs from January 2018 to December 2020.

One of the main challenges was navigating the complex world of tweet data, which included text, photos, and company information. A multimodal strategy was used to solve this, using transformer models for natural language problems and OpenAI CLIP for image processing.
Using **OpenAI's CLIP** (Contrastive Language-Image Pre-training), we implemented a *model* that aids in textual description extraction from images and videos. It demonstrates the model's capacity to understand and describe both images and videos by processing a video by sampling frames and producing subtitles. The generated captions show how flexible the CLIP model is in comprehending and contextualising multimedia data by offering textual representations for the visual material. But because of the significant computing overhead, we switched to a different strategy, putting efficiency first for the later phases.

Although integrating outside resources like

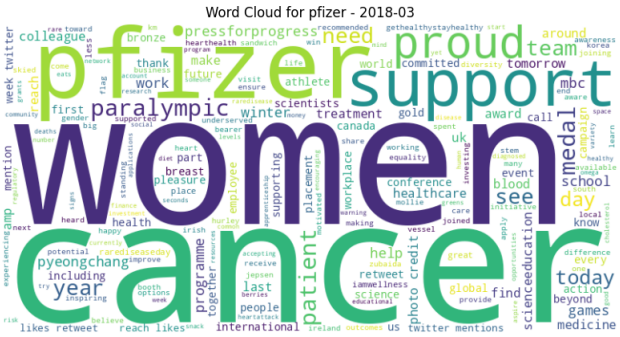


Figure 1: Word cloud using Wikidata

Wikidata increased complexity, it also greatly improved the data's contextual comprehension. We were able to improve embeddings by adding

domain-specific information to company and brand names with the assistance of Wikidata. First, we took clean, tokenized content out of tweets. We then created word clouds for each month and for certain organisations, which helped us identify terms that were used often over time. We obtained information about the industry, official website, and years of operation for entities linked to the deduced company names by using **Wikidata's SPARQL queries**. To better comprehend the brands and businesses referenced in the Twitter data, we also did entity searches to find descriptions, domains, industries, and years active information.

In order to extract the tweet text data, we also created a *program* that entails cleaning the text by eliminating placeholders and extracting hashtags. Emojis are translated into descriptive text. Using spaCy's pre-trained word vectors, text embeddings are produced, resulting in the creation of a new column called "Text_Embeddings." The model's input attributes for imitating tweet content were these embeddings. In order to improve prediction accuracy, a thorough *emoji dataset from Kaggle* was used to forecast the presence of new emojis in the test data. In order to enable the extraction of significant information from content that contains a lot of emojis, the model was trained to tokenize emojis and offer descriptive representations.

One method that was taken into consideration was reinforcement learning, in which a simple caption is first produced. After then, task 1 uses this caption to forecast likes. The data is then cycled back to task 2 to create a better caption depending on the outcomes. This approach was eventually dropped, though, because the objective was to produce a caption that accurately reflected the available data rather than to improve the caption for comparison with ground truth. Although this methodology has been abandoned, reinforcement learning is still a promising technique for improving generated captions instead of urgent tasks.

The history of the project includes combining various approaches, optimising models, and resolving issues pertaining to data. In order to refine models and approaches within the dynamic context of this complex issue space, ongoing review and adaptation were essential.

2 Methodology

2.1 Task 1: Behavior simulation

2.1.1 Preprocessing

Utilising a multimodal strategy, we were able to extract valuable insights from the data. First, we converted text and images into 1024-dimensional embeddings using image binding, a meta-model. PCA was used to reduce the dimensionality of these embeddings, producing succinct 500-dimensional representations. In order to capture temporal nuances, we simultaneously executed one-hot encoding on the date and time data, adding specifics like weekdays and months.

Then, to improve the embedding quality, we used TweetBERT, a specialised language representation model that has been pre-trained on a sizable corpus of Twitter data. To improve and fine-tune these embeddings, we ran them through the TweetBERT model. Better embeddings were produced through this iterative process, and the neural network used these as enhanced input features. Using TweetBERT to fine-tune the model allowed for the capture of subtle, Twitter-specific information, which improved the model's ability to predict user engagement with tweet content.

We used a method centred on the month and any related special days to add temporal relevance to our study. Through the use of a monthly dataset that included notable events or occasions, we were able to determine in real time whether a tweet occurred on any unique date. In the event that a match was found, the relevant data was easily incorporated into the tweet prompt. This strategy anticipated that tweets that coincided with important dates would receive more likes and engagement, therefore it sought to capture and capitalise on the potential impact of timely and contextually relevant content.

2.1.2 Approach

Initially, we combined all of the embeddings for input using a *Multilayer Perceptron (MLP)* approach. In an attempt to be innovative, we gave the MLP architecture a fresh twist. First, a yx1 matrix was produced by neural layer processing of the text and image embeddings (1024 each). A zx1

Score	Actual	Predicted
Mean	777.09	17.86
Standard Deviation	5226.57	1427.05

Table 1: **Mean & standard deviation for task-1**

result was obtained concurrently with the introduction of domain embeddings that were aligned horizontally. Subsequently, the date and time were encoded one-hot and combined with this result to provide a 1x1 output that showed the estimated total number of likes for each datapoint. **Refer Figure 3 in appendix**

2.2 Task 2: Content simulation

2.2.1 Preprocessing

We initiated the preprocessing phase by extracting images from the provided dataset links, ensuring a diverse representation for various media types, including GIFs and Videos. Thumbnails were specifically chosen for GIFs and Videos to capture concise visual information.

2.2.2 Approach

Our main strategy was to use the *LLaVA model*, which is a refined chatbot trained on multimodal instruction-following data collected by GPT and based on **LLaMA/Vicuna**. We implemented a few-shot prompting methodology, which makes use of LLaVA's features and enables the model to generalise from a tiny sample size. By taking into account the distinctive qualities of the data points, such as brand names, domain names, dates, and times, this technique allowed the model to provide appropriate captions for a variety of photos. In order to ensure that the generated captions accurately captured the subtleties of the provided content, we meticulously developed prompts that incorporated the contextual information from the dataset. Our goal was to generate accurate and informative twitter descriptions for the photographs by combining these methodologies. **Refer Figure 4 in appendix**

3 Results

3.1 Task 1

The model predictions demonstrated a substantial difference from the actual engagement metrics, with a mean predicted value of 484.75 compared to the mean actual engagement of 777.09. The standard deviation of the predicted values (1427.05) also exhibited notable variance from the

Score	ROUGE-1	ROUGE-2	ROUGE-L
Precision	41.38	17.86	37.93
Recall	50	21.74	45.83
F-measure	45.28	19.61	41.51

Table 2: **ROUGE score for task-2**

standard deviation of the actual values (5226.57). The root mean square error (RMSE) between the predicted and actual values was considerable, measuring 5049.02, indicating the model's limited precision in predicting user engagement. **Refer Figure 5 in appendix** for the comparison between the actual and predicted distributions.

3.2 Task 2

3.3 ROUGE Score

The scores provided are from ROUGE evaluation metrics, which were applied to the generated text and the captions provided in the training dataset. These metrics are commonly used to assess the quality of generated text in comparison to reference text. Rouge-1 evaluates unigrams (single words), Rouge-2 evaluates bigrams (consecutive pairs of words) and Rouge-L considers the longest common subsequence of words.

Refer Table 2 for ROUGE scores.

Rouge metrics typically reach their peak around 0.50, signifying a very high-quality output. Since our scores hover around 0.40, it is reasonable to assert that our model effectively captures unigrams, bigrams, and subsequences from the reference tweets. This conclusion is supported by the precision, recall, and F-measure scores.

3.4 BLEU 1-4

BLEU (Bilingual Evaluation Understudy) is a metric for assessing machine-generated text quality. Scores range from 0 to 1, with higher values indicating better similarity to reference texts. Commonly used in tasks like machine translation, BLEU is a key tool in natural language processing. The calculated BLEU score is approximately **0.0828**.

Weights: The weights parameter (0.25, 0.25, 0.25, 0.25) indicates that equal importance is given to unigrams, bigrams, trigrams, and four-grams when computing the BLEU score.

Smoothing: The smoothing function (method1) is applied to handle cases where n-grams in the candidate text may not exactly match those in the reference texts.

4 References

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5 Appendix

Github Repository for Team-40

5.1 Assumptions

Thumbnail Selection for Videos and GIFs: To optimize computational resources, thumbnails were chosen as representatives for videos and GIFs, assuming that they encapsulate essential information for content understanding.

Temporal Changes: The analysis assumes that significant variations in user engagement are primarily associated with changes in months rather than years, considering years to remain relatively stable.

index	Query	Description	Domain	Industry	Years Active
0	Apple	fruit of the apple tree	N/A	N/A	N/A
1	Google	American multinational technology company by alphabet inc.	https://about.google/	{'entity-type': 'item', 'numeric-id': 56611700, 'id': 'Q56611700'}	{'time': '+1998-09-04T00:00:00Z', 'timezone': 0, 'before': 0, 'after': 0, 'precision': 11, 'calendarmodel': 'http://www.wikidata.org/entity/Q1985727'}
2	Microsoft	American multinational technology corporation	https://www.microsoft.com/	{'entity-type': 'item', 'numeric-id': 21157865, 'id': 'Q21157865'}	{'time': '+1975-04-04T00:00:00Z', 'timezone': 0, 'before': 0, 'after': 0, 'precision': 11, 'calendarmodel': 'http://www.wikidata.org/entity/Q1985727'}

Figure 2: **Wikidata**

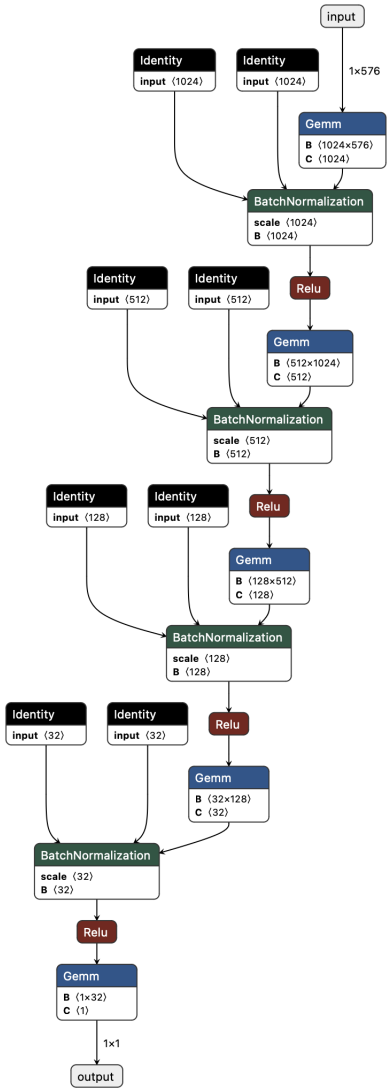


Figure 3: **Multilayer Perceptron**

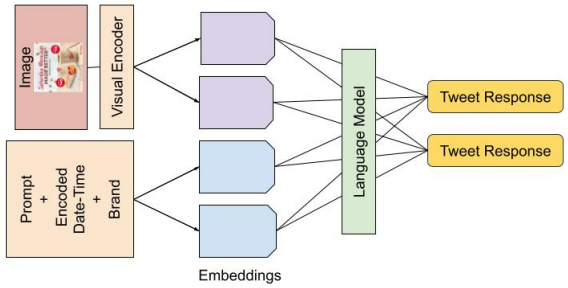


Figure 4: **LLaVA**

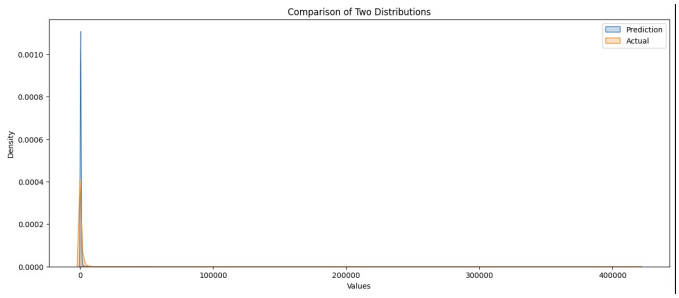


Figure 5: **Comparison between actual and predicted distributions**

```
----------------------------------------------------------------
      Layer (type)               Output Shape         Param #
================================================================
          Linear-1                [-1, 1024]         590,848
     BatchNorm1d-2                [-1, 1024]           2,048
            ReLU-3                [-1, 1024]               0
         Dropout-4                [-1, 1024]               0
          Linear-5                 [-1, 512]         524,800
     BatchNorm1d-6                 [-1, 512]           1,024
            ReLU-7                 [-1, 512]               0
         Dropout-8                 [-1, 512]               0
          Linear-9                 [-1, 128]          65,664
    BatchNorm1d-10                 [-1, 128]             256
           ReLU-11                 [-1, 128]               0
        Dropout-12                 [-1, 128]               0
         Linear-13                  [-1, 32]           4,128
    BatchNorm1d-14                  [-1, 32]              64
           ReLU-15                  [-1, 32]               0
         Linear-16                   [-1, 1]              33
================================================================
Total params: 1,188,865
Trainable params: 1,188,865
Non-trainable params: 0
----------------------------------------------------------------
Input size (MB): 0.00
Forward/backward pass size (MB): 0.05
Params size (MB): 4.54
Estimated Total Size (MB): 4.59
----------------------------------------------------------------
```

Figure 6: **Architecture Summary**